Malaviya National Institute of Technology Jaipur
Department of Computer Science and Engineering

Build Your CryptoLabX Toolkit

Project foundation: structure, CLI, file analysis and logging

Course	Cryptography Laboratory (22CPP307)
Assignment	Assignment 1 (Week 1)
Group	Group 10 (Even group number)
Team members	Rishi Saini (2024UCP1566), Nandini Verma (2024UCP1667)
Instructor	Dr. Meenakshi Tripathi
Submission date	15 October 2026
GitHub repository	https://github.com/the-sage-00/Cryptography-LabX

1. Task checklist

Task	Requirement	Where / status
1	Git repository + first commit with project structure	Done - first commit 'init: set up project structure with all required modules'
2	Folder structure	classical, attacks, math, modern, analysis, datasets, outputs, docs, tests, utils, main.py, README.md, requirements.txt
3	Menu-driven CLI (Encrypt, Decrypt, Attack, Analyze, Exit)	main.py + utils/menu.py (unfinished options print 'Coming Soon')
4	File analysis (characters, words, lines, unique characters, letter frequency)	analysis/file_analyzer.py
5	Log file with date, time and option	utils/logger.py → outputs/cryptolabx.log
6	At least five dataset files	datasets/: 5 text files
7	README (project, team, structure, future modules)	README.md

2. Project structure

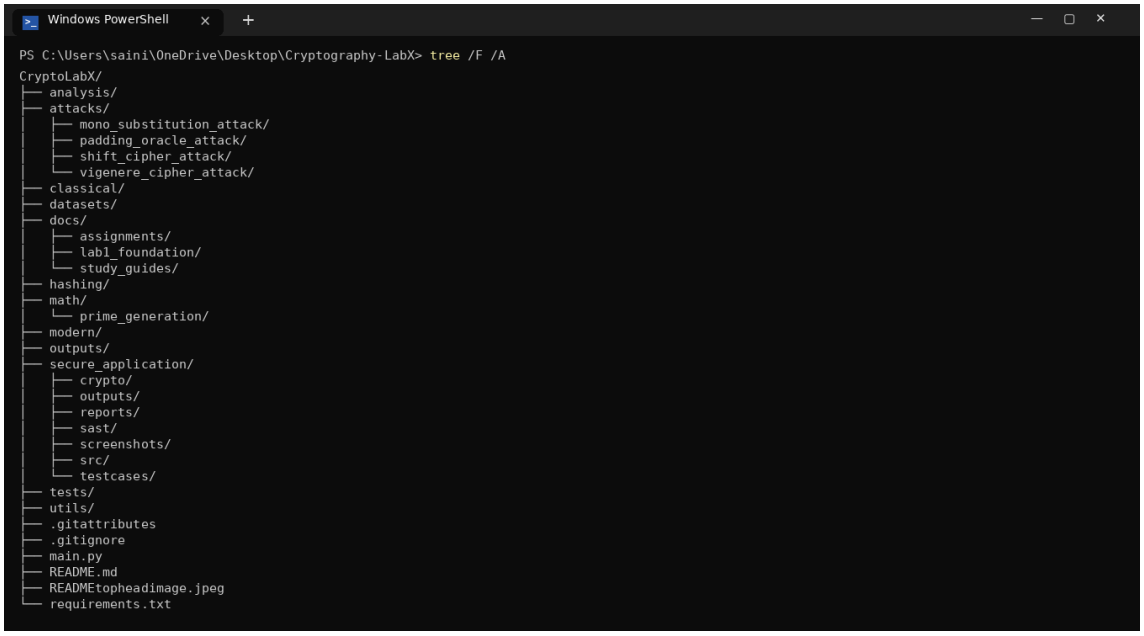


Figure 1 - Project structure

3. Working menu

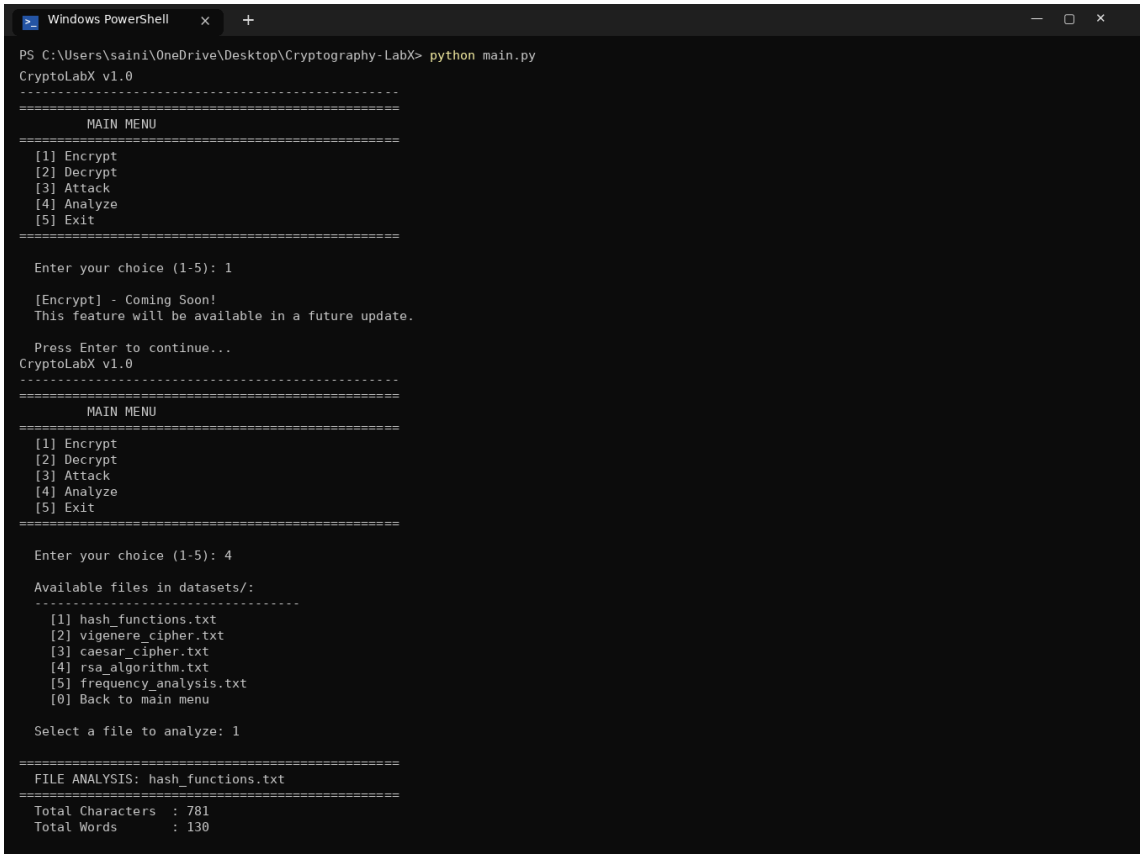


Figure 2 - Main menu; options 1-3 report 'Coming Soon'

4. File analysis output

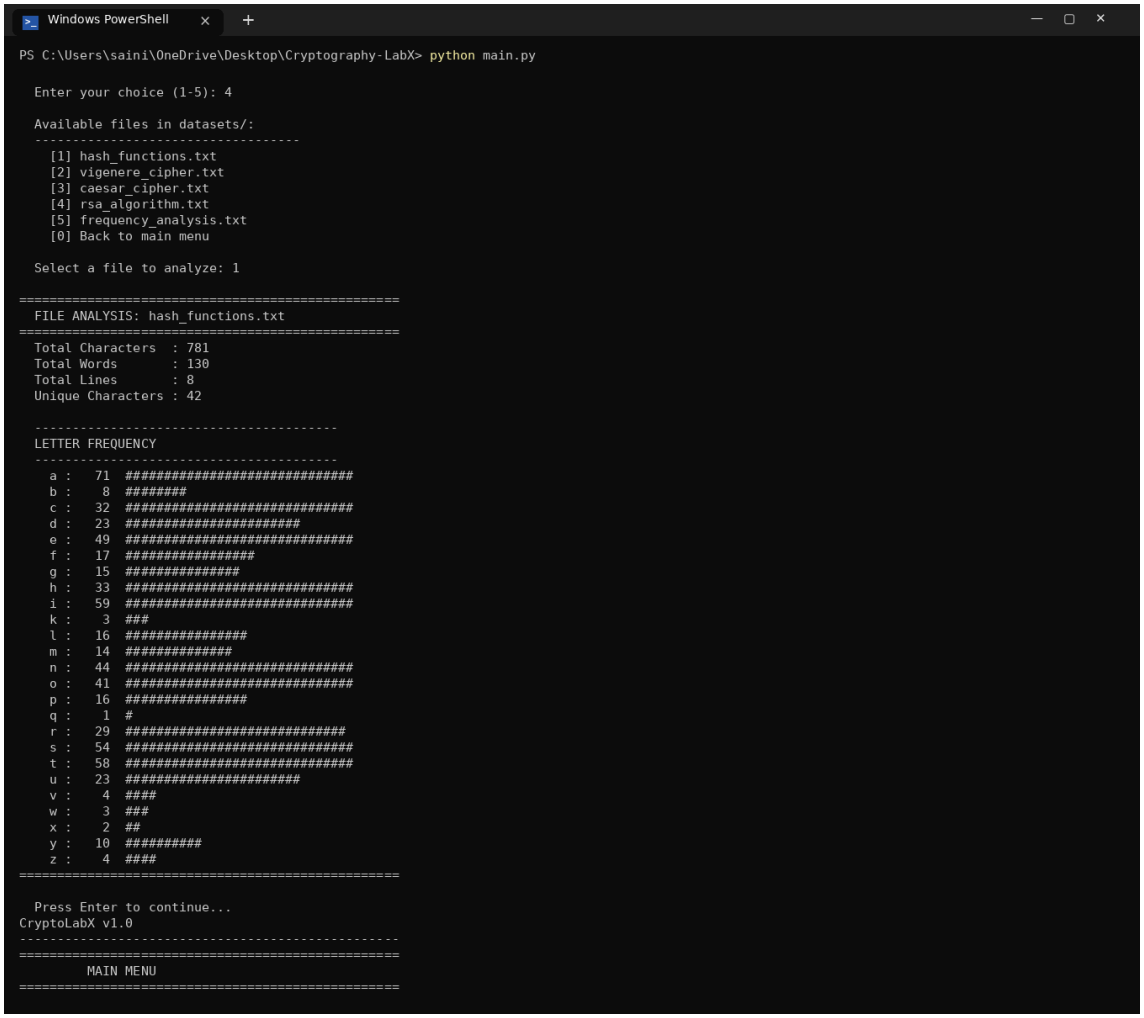


Figure 3 - Option 4 (Analyze) on a dataset file

5. Generated log file

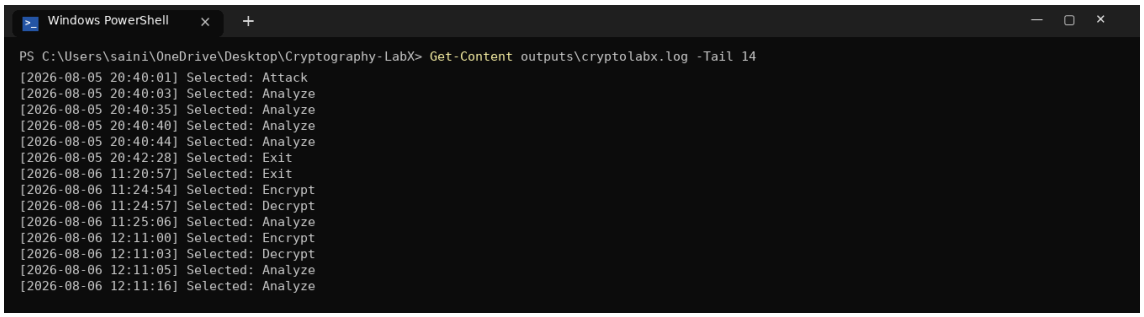


Figure 4 - outputs/cryptolabx.log (timestamp and selected option per run)

6. Design notes

- Modular packages (utils, analysis) keep the CLI thin; every later assignment is added under attacks/, math/, modern/ without touching main.py.
- The logger appends one line per selection, so the log is a permanent audit trail of toolkit use.
- No cryptographic algorithm is implemented in this assignment, as required.